

5 (**short**) An air compressor operates in steady flow with air entering with pressure p_1 and temperature T_1 , and leaving at pressure p_2 . Within the compressor the air undergoes a reversible polytropic process, $pv^n = \text{constant}$, where v is the specific volume and n is a constant. Assume air behaves as a perfect gas and neglect changes in kinetic and potential energy between the inlet and the outlet.

(a) Starting from the expression $dw_x = -v dp$ for a reversible steady flow, show that w_x (the shaft work transfer from the compressor per unit mass of air) is given by

$$w_x = -\frac{nR}{n-1}(T_2 - T_1)$$

where R is the gas constant per unit mass of air. [5]

(b) Derive an expression for the heat transfer to the compressor per unit mass of air in terms of R , T_1 , T_2 , n and the ratio of the specific heat capacities γ . [5]